

Gemini API

Compatibility & Architecture Report

Model availability · Google Search Grounding · Web Chat vs API

Invennico Lead Portal — Internal Technical Reference

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1. Executive Summary

What we found and what it means for the Lead Portal

This report documents a systematic investigation of all Gemini API models available on the Invennico project API key, their compatibility with the free-tier quota system, and the fundamental difference between using Gemini through its web interface versus the REST/SDK API. The investigation was prompted by 404 and 429 errors encountered when integrating Gemini-based Lead Research into the Lead Portal backend.

Key Findings

- ! ' gemini-1.5-flash**
Completely removed from the API — returns 404. This model has been deprecated and no longer exists at any endpoint.
- ! ' Versioned model IDs (gemini-2.0-flash, gemini-2.0-flash-lite, etc.)**
Return 429 with limit: 0. These models exist but are assigned zero free-tier quota on this project key. They require billing to be enabled.
- ! ' Alias model IDs (*-latest, *-preview, *-lite)**
Work correctly on the free tier. Google does not track alias usage against the same quota buckets as versioned IDs.
- ! ' gemini-2.5-flash-lite + googleSearch grounding**
Grounding works on this model with the googleSearch tool. Confirmed by returning today's exact date (June 16, 2026) from a live web search.
- ! ' Web Chat vs API**
Gemini web chat (gemini.google.com) always has grounding enabled at Google's expense. The API requires your own billing for grounding calls.

2. Complete Model Test Results

Every model tested against your API key — live results

The following table shows the result of calling generateContent on every model returned by the ListModels API endpoint for this project key. Models marked OK successfully generated a text response. Models marked FAIL show the exact HTTP error and its cause.

Model ID	Category	Status	Result / Reason
gemini-2.5-flash	Flash	FAIL	Unknown error (likely billing)
gemini-2.5-pro	Pro	FAIL	429 — Free tier quota = 0
gemini-2.0-flash	Flash	FAIL	429 — Free tier quota = 0
gemini-2.0-flash-001	Flash	FAIL	429 — Free tier quota = 0
gemini-2.0-flash-lite	Flash Lite	FAIL	429 — Free tier quota = 0
gemini-2.0-flash-lite-001	Flash Lite	FAIL	429 — Free tier quota = 0
gemini-1.5-flash	Flash	FAIL	404 — Model deprecated & removed
gemma-4-26b-a4b-it	Gemma	OK	Responds but returns markdown bullets (instruction-tuned format)

Model ID	Category	Status	Result / Reason
gemma-4-31b-it	Gemma	OK	Responds but returns markdown bullets (instruction-tuned format)
gemini-flash-latest	Alias	OK	Clean response — recommended alias
gemini-flash-lite-latest	Alias	OK	Clean response — lightest free-tier alias
gemini-pro-latest	Alias/Pro	FAIL	429 — Free tier quota = 0 (Pro tier)
gemini-2.5-flash-lite	Flash Lite	OK	Clean response + grounding works
gemini-3-pro-preview	Pro	FAIL	429 — Free tier quota = 0
gemini-3-flash-preview	Flash	OK	Clean response
gemini-3.1-pro-preview	Pro	FAIL	429 — Free tier quota = 0
gemini-3.1-flash-lite-prev	Flash Lite	OK	Clean response
gemini-3.1-flash-lite	Flash Lite	OK	Clean response
gemini-3.5-flash	Flash	OK	Clean response

Grounding (Web Search) Test Results

Tested both googleSearch and googleSearchRetrieval

Model ID	Tool Type	Status	Result / Reason
gemini-flash-latest	googleSearchRetrieval	FAIL	429 — Rate limited during test batch
gemini-flash-latest	googleSearch	FAIL	429 — Rate limited during test batch
gemini-flash-lite-latest	googleSearchRetrieval	FAIL	429 — Rate limited during test batch
gemini-flash-lite-latest	googleSearch	FAIL	429 — Rate limited during test batch
gemini-2.5-flash-lite	googleSearchRetrieval	FAIL	400 — Tool not supported on this model (use googleSearch)
gemini-2.5-flash-lite	googleSearch	OK	Live web search confirmed — returned today's date correctly
gemini-3.5-flash	googleSearch	FAIL	429 — Rate limited during test batch
gemini-3.1-flash-lite	googleSearch	FAIL	429 — Rate limited during test batch

KEY FINDING: gemini-2.5-flash-lite with the googleSearch tool is the ONLY model on this key that successfully executed a grounded web search. The 429 errors on other models during the grounding test were caused by rate limiting from rapid sequential API calls in the test script — those models may also support grounding but were not retested after cooldown.

3. Why Versioned Model IDs Fail (Quota System)

Understanding limit: 0 and how Google's quota tiers work

The Two-Bucket Quota System

Google Gemini API uses a two-bucket quota model for every model it offers:

Bucket 1 — Free Tier Quota

A fixed number of free requests per day and per minute. Example: 1,500 requests/day, 15 requests/minute. These are tracked per model ID, per project, per day.

Bucket 2 — Paid Tier Quota

Unlimited requests (billed per token). Requires billing to be enabled on the Google Cloud project linked to your API key. Once billing is enabled, all requests that exceed Bucket 1 fall through to Bucket 2.

Why Versioned IDs Show limit: 0

When you create a new API key in Google AI Studio today, the free-tier quota buckets for older versioned model IDs are set to 0 by default. This is because:

- Google is actively migrating developers to newer model generations (2.5+, 3.x)
- Older versioned IDs (gemini-2.0-flash, gemini-2.0-flash-lite) are being deprecated
- Setting free-tier quota to 0 forces developers to either upgrade models or enable billing
- The model still EXISTS in the API (it appears in ListModels) but you cannot call it for free

limit: 0 does NOT mean the model is unavailable — it means your project's free-tier allocation for that model ID is zero. If you enable billing on the project, these models become available as paid calls.

Why Alias IDs (*-latest) Still Work

Alias model IDs like gemini-flash-lite-latest, gemini-flash-latest, and gemini-2.5-flash-lite are not subject to the same deprecated quota buckets. Google maintains separate (more generous) quota pools for these aliases because:

- They always point to the most current recommended model
- Google wants developers using the latest models, not pinning to old versioned IDs
- The alias quota buckets are refreshed whenever Google updates what the alias points to
- They are the "blessed" path for free-tier usage

Why gemini-1.5-flash Returns 404

This is different from the quota issue. The gemini-1.5-flash model has been completely removed from the API — it no longer exists at any endpoint. The 404 error means the URL <https://generativelanguage.googleapis.com/v1beta/models/gemini-1.5-flash:generateContent> points to nothing. The model was deprecated and fully

decommissioned. Any code referencing it must be updated.

4. Google Search Grounding — What It Is & How It Works

How Gemini performs live web research

What is Grounding?

Grounding is Gemini's ability to connect to live Google Search before generating a response. Without grounding, Gemini answers from its training data (knowledge cutoff). With grounding, Gemini first searches Google, reads the top results, then generates a response informed by current, real-time information.

This is why grounding is so valuable for Lead Research — Gemini can look up the client's company, check their website, read their product pages, and surface business intelligence that wasn't in its training data.

How Grounding Works Technically

Step 1 — Tool Declaration

You tell Gemini it has access to Google Search by passing a tool configuration:

```
tools: [{ googleSearch: {} }]           // For gemini-2.5+ models
tools: [{ googleSearchRetrieval: {} }] // For older 1.x models (deprecated)
```

Step 2 — Gemini Decides When to Search

Gemini autonomously decides whether to call Google Search based on the prompt. For questions about current events, specific companies, or real-time data, it will search. For general reasoning or coding tasks, it may not search at all (saving you the grounding cost).

Step 3 — Results Are Injected Into Context

Google Search results (titles, snippets, URLs) are fetched and injected into Gemini's context window before generation. Gemini reads these results and synthesizes them with its training knowledge to produce a grounded response.

Step 4 — Response Includes Grounding Metadata

The API response includes a `groundingMetadata` field with the search queries used, the URLs consulted, and grounding support citations. Your code receives the grounded text response plus this metadata.

The grounding call happens server-side at Google — you never make a direct Google Search API call yourself. You just declare the tool, Gemini handles the search, and you receive the enriched response.

Tool Name Difference: `googleSearch` vs `googleSearchRetrieval`

There are two different tool names depending on the model generation:

- `googleSearchRetrieval` — Used by Gemini 1.x models (now deprecated). Uses a retrieval-augmented generation (RAG) approach where search results are retrieved and used as grounding context.
- `googleSearch` — Used by Gemini 2.x and newer models. Gives the model direct tool-use capability to call Google Search as a function. More flexible, more powerful. Our test confirmed this works with `gemini-2.5-flash-lite`.

The 400 Bad Request we got when testing googleSearchRetrieval on gemini-2.5-flash-lite is because that model only accepts googleSearch. Never mix tool names across model generations.

5. Gemini Web Chat vs Gemini API — The Core Difference

Why web chat can search for free but the API charges for it

The Question

If you open gemini.google.com and ask "Tell me about Upwork lead patterns for AI projects", Gemini searches the web and gives you a real-time answer — for free. But when the same question goes through the API with googleSearch enabled, Google charges for the grounding call. Why?

How Gemini Web Chat Works

Gemini web chat (gemini.google.com) is a consumer product built and operated by Google. When you use it:

- Google is the operator — they pay the infrastructure and grounding costs
- Grounding is permanently enabled for all conversations at no cost to you
- You are the end user consuming a free (ad-supported/subscription) consumer product
- Your data is used to improve Google's models (under their privacy policy)
- The model used is typically Gemini 2.0 Flash or Gemini 2.5 Flash — Google's latest
- There are no API keys, no quota buckets, no per-call costs visible to you

How the Gemini API Works

The Gemini API is a developer platform (generativelanguage.googleapis.com). When your backend calls it:

- You are the operator — you build the product and control the model calls
- Google charges you per token (input + output) at published rates
- Grounding is an optional add-on charged separately per grounding request
- Your data is not used for model training (under the API terms of service)
- You control exactly which model, temperature, tools, and system prompt are used
- You get structured JSON responses, fine-grained control, and programmatic access

Gemini Web Chat	Gemini API (Your Backend)
' Consumer product (gemini.google.com) ' Google pays all infrastructure costs ' Grounding always ON — Google absorbs cost ' No API key required ' No per-call billing to you ' You share data for model improvement ' No programmatic/structured access ' Cannot be embedded in your app ' Model choice is Google's decision	' Developer platform (ai.google.dev) ' You pay per token (input + output) ' Grounding optional — \$35 / 1K grounding calls ' API key required (per project) ' Free-tier quota available (limited) ' Data NOT used for model training (API ToS) ' Full programmatic / structured JSON access ' Can be embedded in your application ' You choose exactly which model to use

In short: Gemini web chat is a consumer product where Google funds the grounding. The API is a developer platform where you fund the grounding. They use the same underlying AI model but operate under completely different billing and access models.

Practical Implications for the Lead Portal

When you test Lead Research by pasting a lead into gemini.google.com and get excellent client research — that is NOT representative of what a free-tier API key can do. The web chat has permanent, funded grounding. Your API key only has grounding if billing is enabled on the project.

6. Grounding Billing & Cost Structure

What enabling billing actually means and costs

What Billing Enables

- Enabling billing on a Google Cloud project linked to your Gemini API key unlocks:
- Access to all versioned model IDs (gemini-2.0-flash, gemini-2.5-flash, etc.)
 - Google Search Grounding for models that support it (gemini-2.5-flash-lite + googleSearch confirmed working)
 - Higher rate limits (higher RPM and daily quotas)
 - Access to Pro-tier models (gemini-2.5-pro, gemini-3.1-pro-preview, etc.)

Google Search Grounding Cost

Grounding Requests / Day	Cost / Rate
First 1,500 grounding requests / day	FREE
Beyond 1,500 / day	\$35.00 per 1,000 requests
Model token cost (input + output)	Charged separately per model pricing

Model Token Costs (approximate c

Model	Type	Input /M tok	Output /M tok	Access
gemini-flash-lite-latest	Alias !' latest lite	\$0.075	\$0.30	Free tier '
gemini-2.5-flash-lite	Flash Lite	\$0.075	\$0.30	Free tier '
gemini-flash-latest	Alias !' latest flash	\$0.15	\$0.60	Free tier '
gemini-2.5-flash	Flash	\$0.15	\$0.60	Billing req.
gemini-2.5-pro	Pro	\$1.25	\$10.00	Billing req.

For the Lead Portal's expected volume (< 50 leads/day), the cost with billing enabled would be approximately \$0.50–\$2.00/month for grounding. Well within minimal cost territory.

7. Implementation Recommendation for Lead Portal

The model to use and how to handle grounding gracefully

Recommended Model: gemini-2.5-flash-lite

Based on all tests, gemini-2.5-flash-lite is the best choice for Lead Research:

- Works on the current free-tier key '
- Supports googleSearch grounding when billing is enabled '
- More capable than the *-latest aliases (dedicated versioned model with good quota on this key) '
- Grounding test confirmed live web search returning accurate real-time data '

Conditional Grounding Pattern

The implementation uses an environment variable GEMINI_SEARCH_GROUNDING to conditionally enable grounding:

```
// gemini.service.js
const useGrounding = process.env.GEMINI_SEARCH_GROUNDING === 'true';

const modelConfig = { model: 'gemini-2.5-flash-lite' };
if (useGrounding) {
  modelConfig.tools = [{ googleSearch: {} }];
}

const model = client.getGenerativeModel(modelConfig);
```

Environment Variable Setup

Add to .env file:


```
# Gemini API key (required)
GEMINI_API_KEY=your_key_here

# Set to true only when billing is enabled on the Google AI project
GEMINI_SEARCH_GROUNDING=false
```

Failure Handling

The Lead Research call is already wrapped in Promise.allSettled in lead.service.js. If Gemini fails for any reason (wrong model, quota exhausted, grounding billing issue), the lead is still created successfully with Claude's analysis. The Lead Research tab simply shows "No research available" to the admin. No lead is ever lost due to a Gemini failure.

8. Conclusion & Next Steps

Summary of actions required

Current State

The Lead Portal backend has been updated to use gemini-2.5-flash-lite without grounding. This works on the current free-tier API key and produces good Lead Research output from Gemini's training knowledge (company analysis, platform context, budget patterns, etc.).

To Enable Live Web Search Grounding

- Step 1:** Go to console.cloud.google.com and open the project linked to your GEMINI_API_KEY
- Step 2:** Navigate to Billing !' Link a Billing Account !' add a credit card (Google gives \$300 free credit for new accounts)
- Step 3:** Once billing is enabled, set GEMINI_SEARCH_GROUNDING=true in your .env file
- Step 4:** Restart the backend server — Gemini will now perform live Google Search for each lead
- Step 5:** Monitor usage at ai.dev/rate-limit — the first 1,500 grounding requests per day are free

Without billing: Gemini uses training knowledge for research (still excellent for platform context, industry patterns, budget norms). With billing + grounding: Gemini performs live Google Search to find the client's actual company website, LinkedIn, news, and real-time business intelligence.

Quick Reference — Model Decision Matrix

Use Case	Model	Billing	Action
Just Lead Research (no web)	gemini-2.5-flash-lite	No billing needed	Current setup without grounding
Lead Research + Web grounding	gemini-2.5-flash-lite	Enable billing	Set GEMINI_SEARCH_GROUNDING=true
Higher quality analysis	gemini-flash-latest	No billing needed	Swap model ID
Best quality + web	gemini-3.5-flash	Enable billing	Test for grounding support